



100 Quantum Bits Computer

A startup known as Atom computing uses optical tweezers to trap the QBits in a vacuum chamber. <http://digit.in/aug21-53>

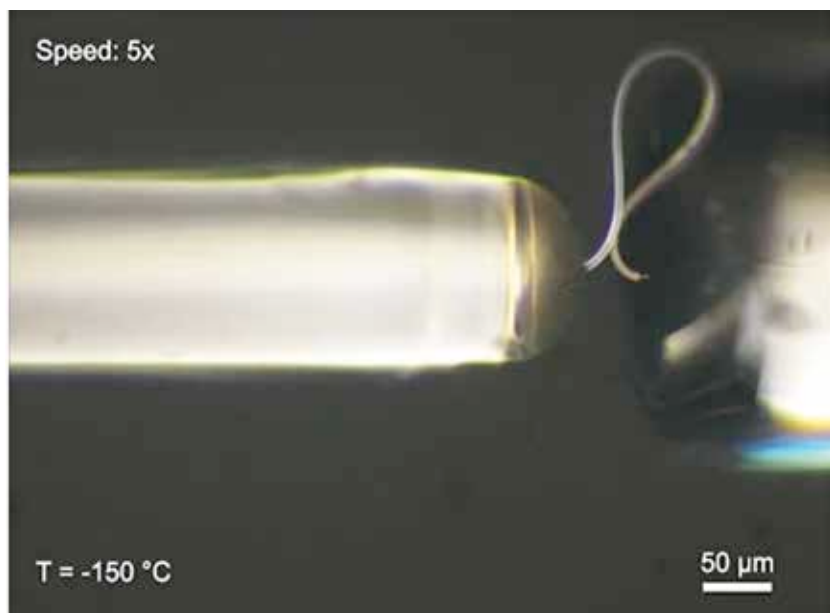


Holy grail of energy

Helion energy's facility will use zero-carbon non-ignition fusion technology to produce affordable power. <http://digit.in/aug21-54>



It is like glass but made of water



It almost formed a full circle

single pure crystal that forms the ice microfibres. These tiny fibres, when stressed, also flex and not break.

After the formations of these crystals, the team of scientists used microscopic tools to push the fibres from both ends, demonstrating their flexibility. Instead of snapping into half as an icicle would, the microfiber bent and snapped back to its original state once the force from either of the ends was released. What was even more fascinating was that after snapping back to their original form, these crystals did not have any permanent deformity in their structure, allowing multiple bendings of the fibre once it was made. These fibres behave much

like optical fibres, and got increasingly flexible and springy as the temperatures were reduced. At -70 degrees celsius the fibres could be bent around to form circular loops. The observed strains were close to the theoretical limits of how elastic ice can possibly get.

These ice microfibers have very suitable optical characteristics, similar to the optical fibers made out of silica or glass. The team that created the ice microfibres previously had experience in creating optical microfibres. Considering many quantum computers and supercomputers are operated at even lower temperatures than the chamber in

which the ice microfibres are created, the low temperature does not seem to be a constraint on future computing applications. The fibres can replace state of the art waveguides on chips. According to the researchers, these structures were capable of transmitting light through them, with less than 1 per cent loss in energy. The ice can potentially be used as sensors for studying air pollution in an area. If these fibres are grown from water found in a particular area, then the impurities could be easily discovered by studying lit up ice microfibres.

There are about 20 crystalline phases of ice and even an amorphous phase of ice, and the creation of these ice microfibres could provide researchers with a new tool for studying the phase transitions. In the experiment, ice moved from the regular crystalline ice found widely in nature, known as Ih to a phase known as ice II when bent, which is a rhomboid crystalline form. These different phases of ice are formed under various temperature and pressure conditions, have different ways in which the ice crystals are ordered. This is why this discovery could just be the beginning of an increased understanding of ice, opening up even more applications in the future. **[1]**



Look how thin the fibre is